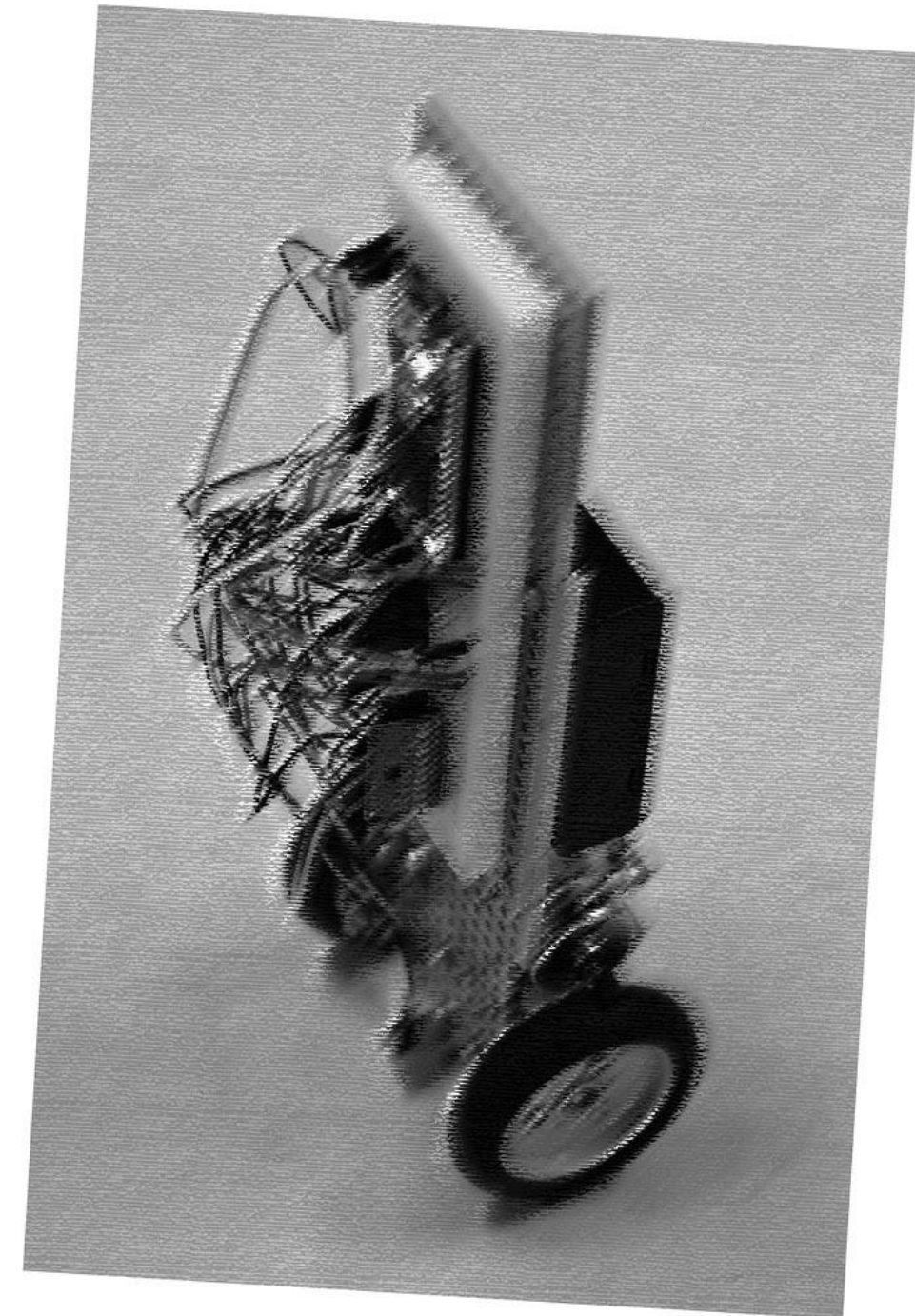


PRACTICAL EXERCISE · WEEK 2

Construction of a Wheeled Inverted Pendulum

| Shaoqi (シャオ チー)

COURSE	Department of Precision Engineering · 2026 S1S2
SESSIONS	Mondays, 13:00–18:00 · Room 326, Engineering Bldg. 14
TODAY	Course Introduction — 10 minutes



THE FOUR-WEEK ARC

Timeline

1

Week 1 Standing

Assembly & bring-up
Angle-only **standing** demo

2

Week 2 Position control

Build a rotary encoder
Position control
Simulation in Simulink

3

Week 3 EXTENSION

Load & disturbance tests
Advanced control (LQR/MPC)
.....

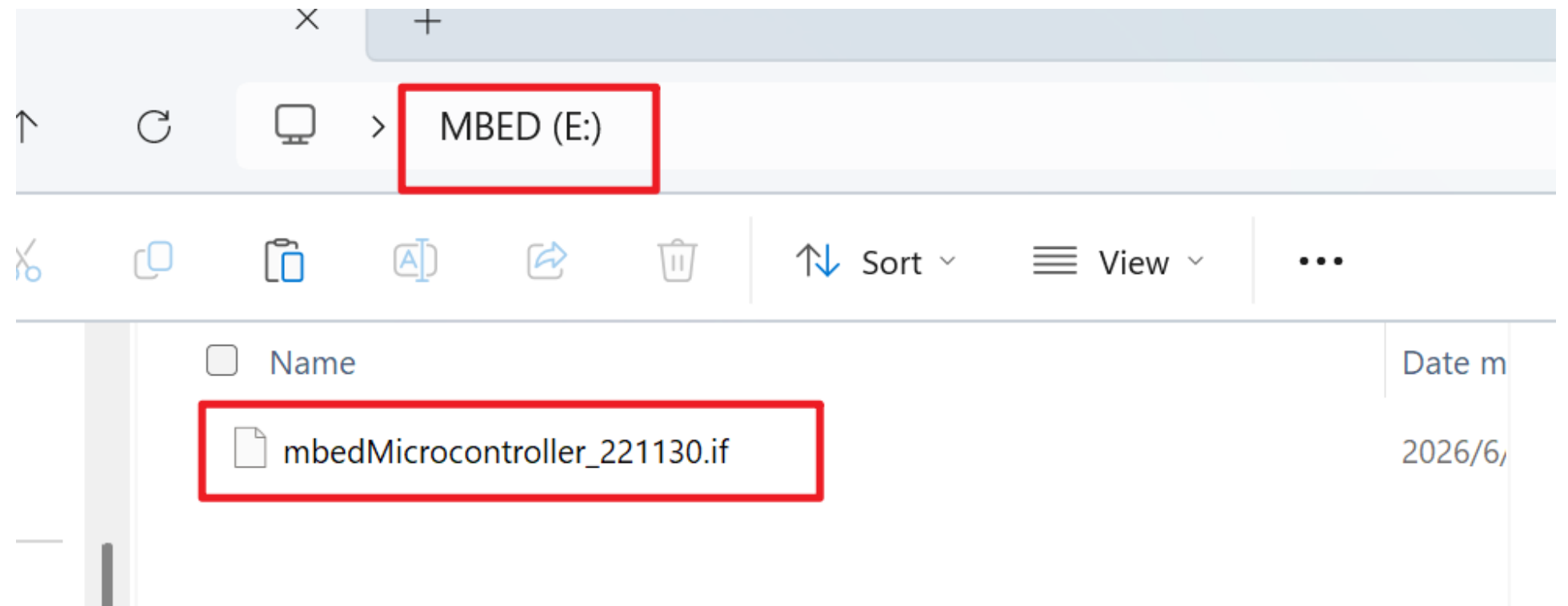
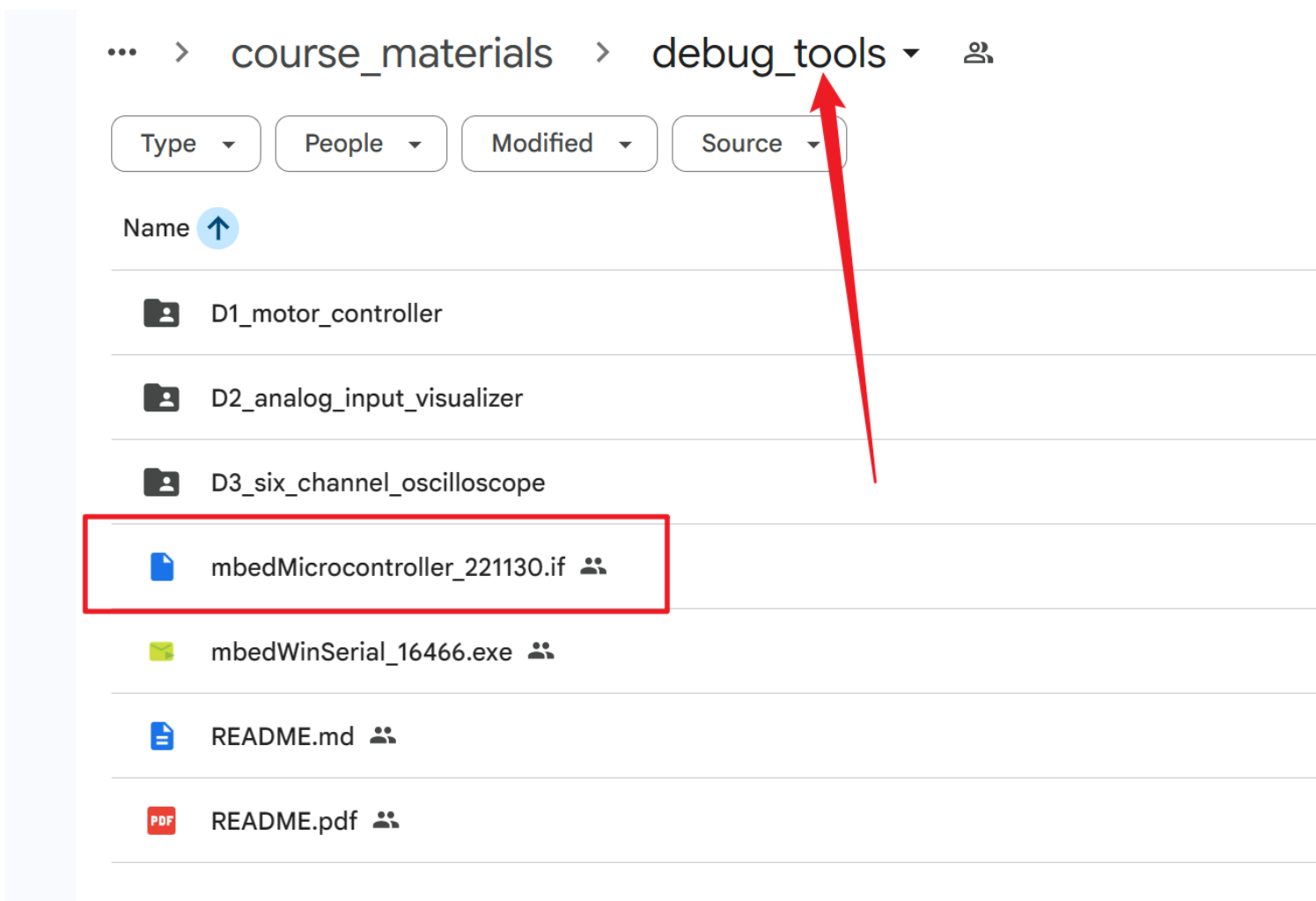
4

Week 4 presentation

10-minute presentation
Live demo & Q&A
Final grade · 40%

Week 3 is for extended tasks, or you can continue working on the tasks of Week 1 and Week2

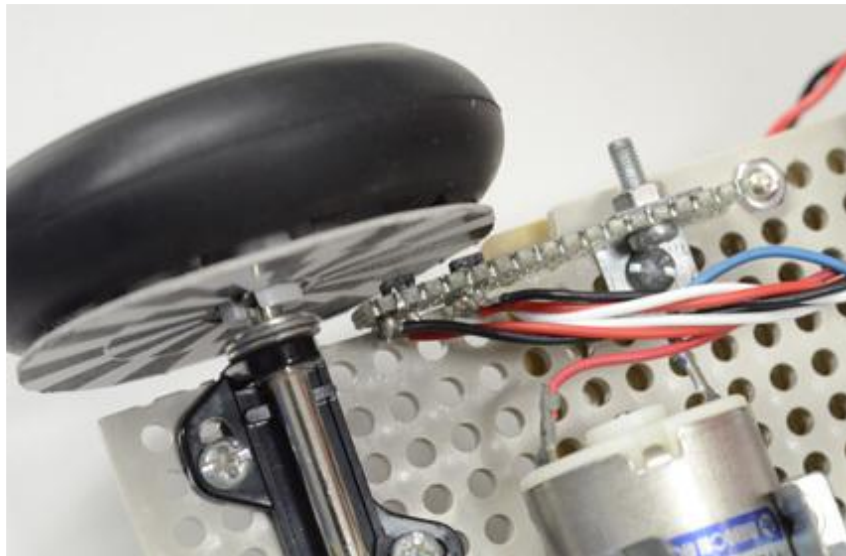
If you are unable to connect to the Windows serial port, you can try updating the mbed firmware.



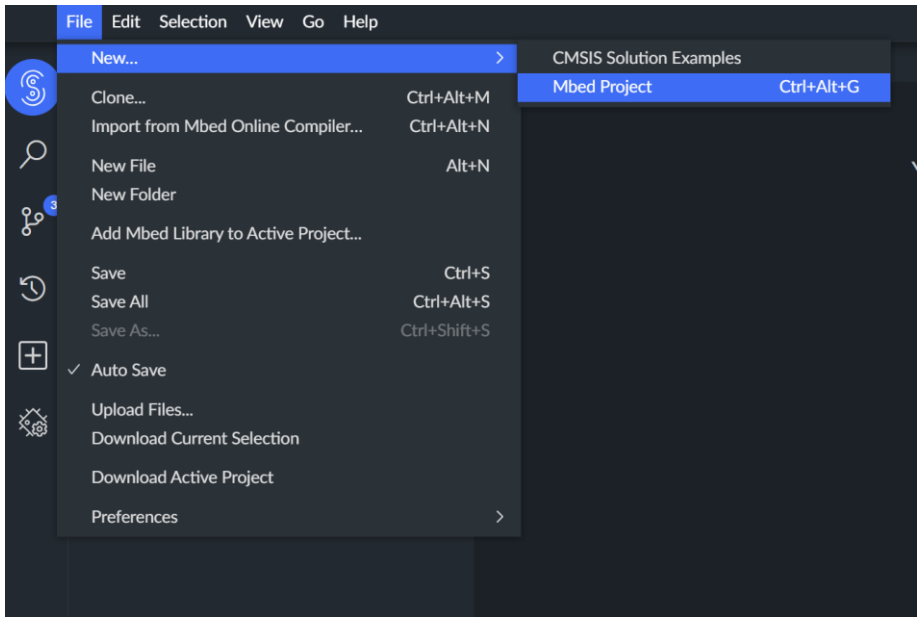
Week 2

Assign the tasks among yourselves.

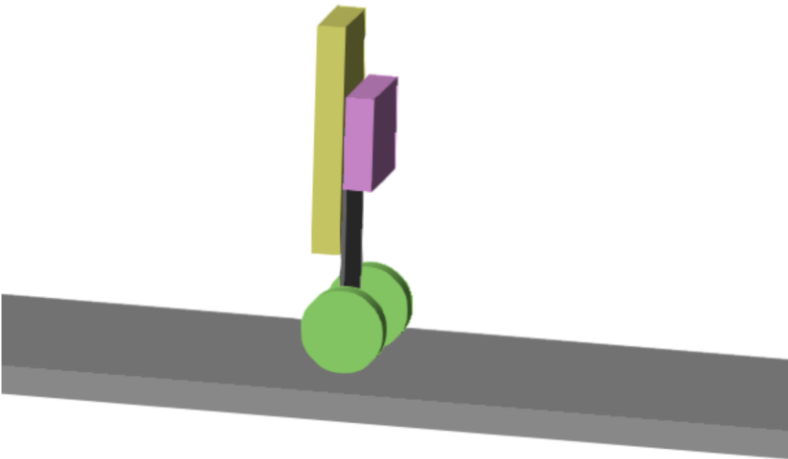
A TASK A
STUDENT A · Hardware
Encoder Fabrication

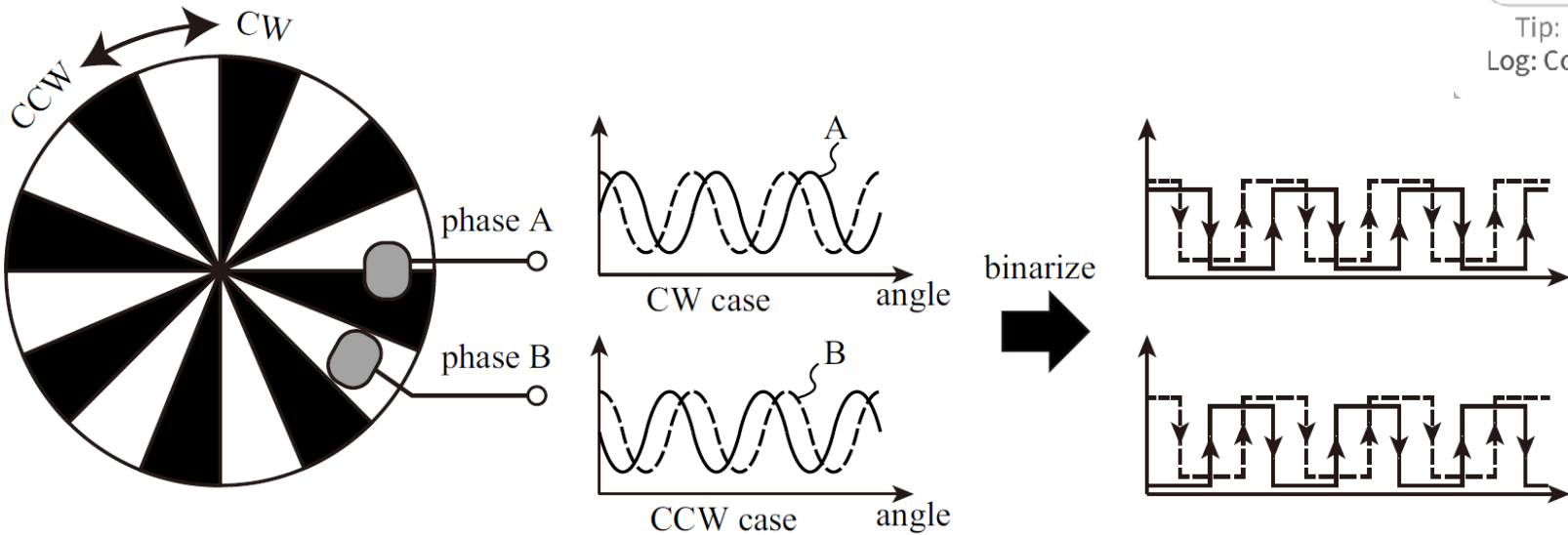
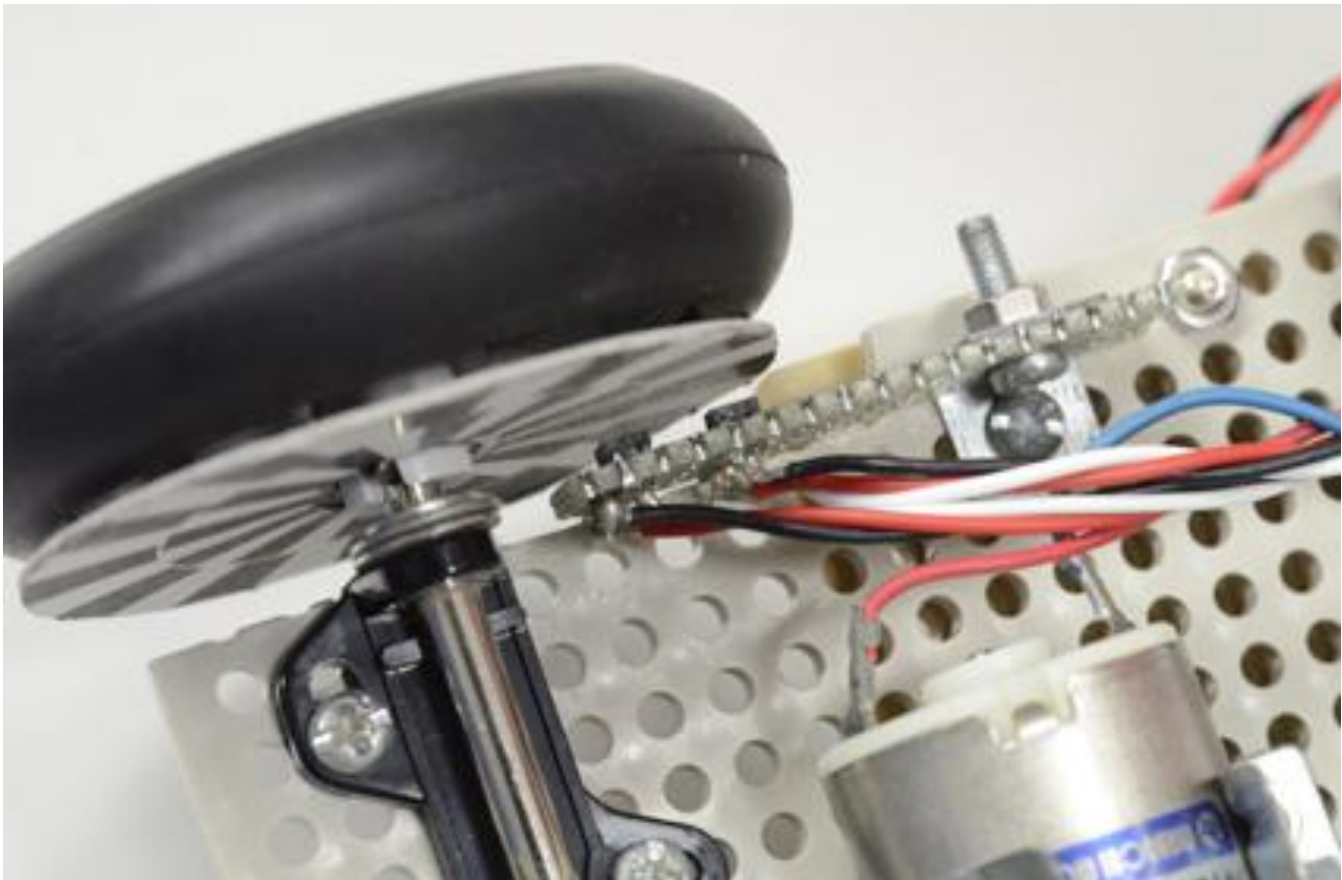


B TASK B
STUDENT B · FIRMWARE
**Add Encoder Feedback to
Microcontroller**



C TASK C
STUDENT C · SIMULATION
Simulink Simulation





analog_input_visualizer

4ch Analog Debug (Frame: AA 55 CH1 CH2 CH3 CH4)

Port: COM3 (click to cycle)

Refresh

Disconnect

Decoded Channels (raw 0..255) [CH1=p16, CH2=p17, CH3=p19, CH4=p20]

CH1 (p16): 79	CH2 (p17): 62	CH3 (p19): 151	CH4 (p20): 10
CH1: 0.310	CH2: 0.243	CH3: 0.592	CH4: 0.039

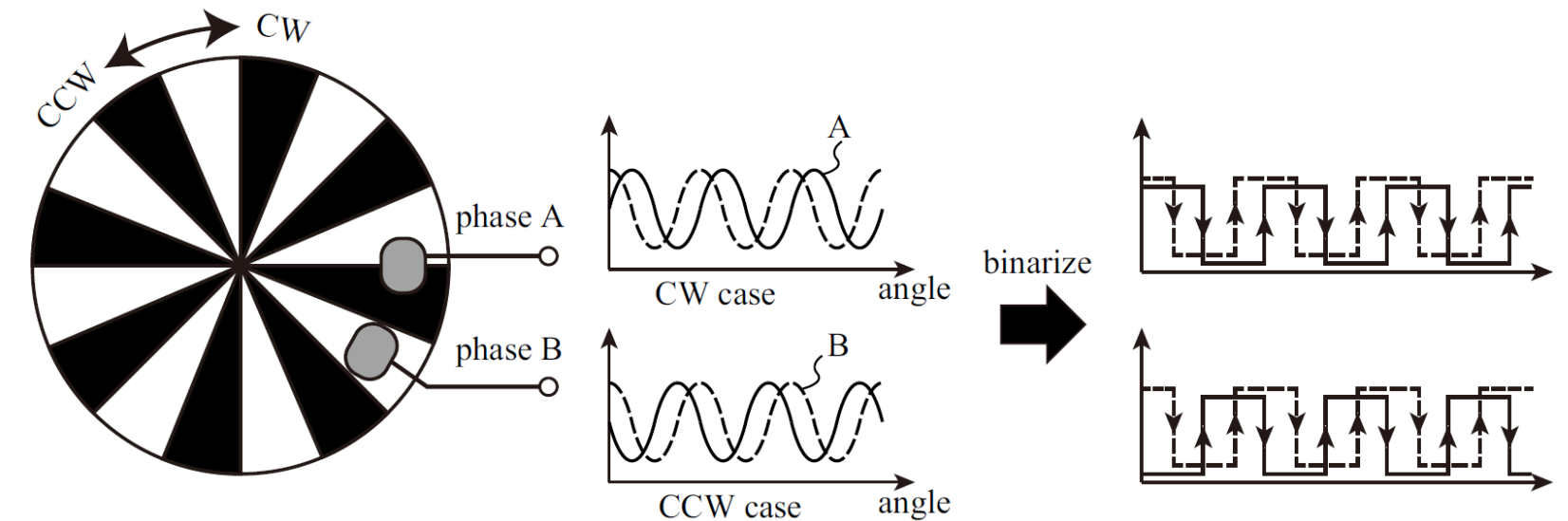
Frames: 897 Bad bytes: 0 (auto re-sync by header)

Simple Bar View (0..1)

CH1 p16		0.310
CH2 p17		0.243
CH3 p19		0.592
CH4 p20		0.039

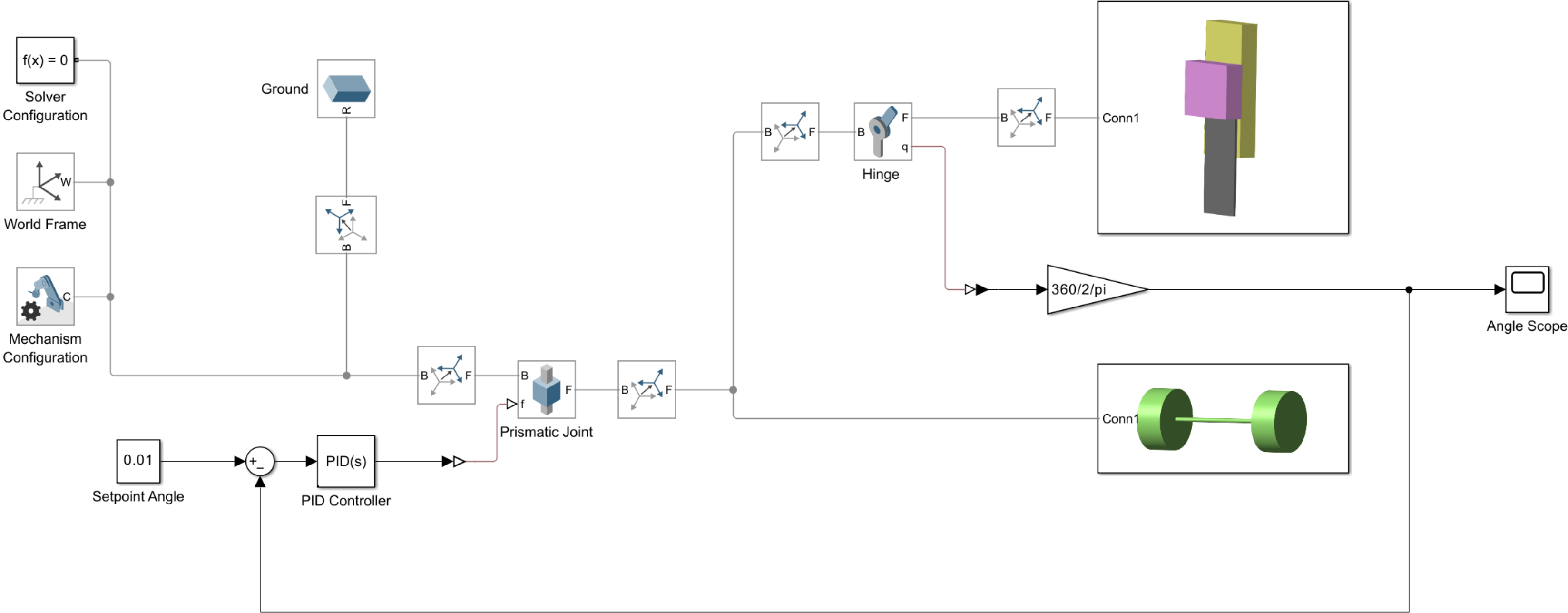
Tip: if values look frozen -> check baud / wiring / device is sending frames.
Log: Connected: COM3 @ 115200

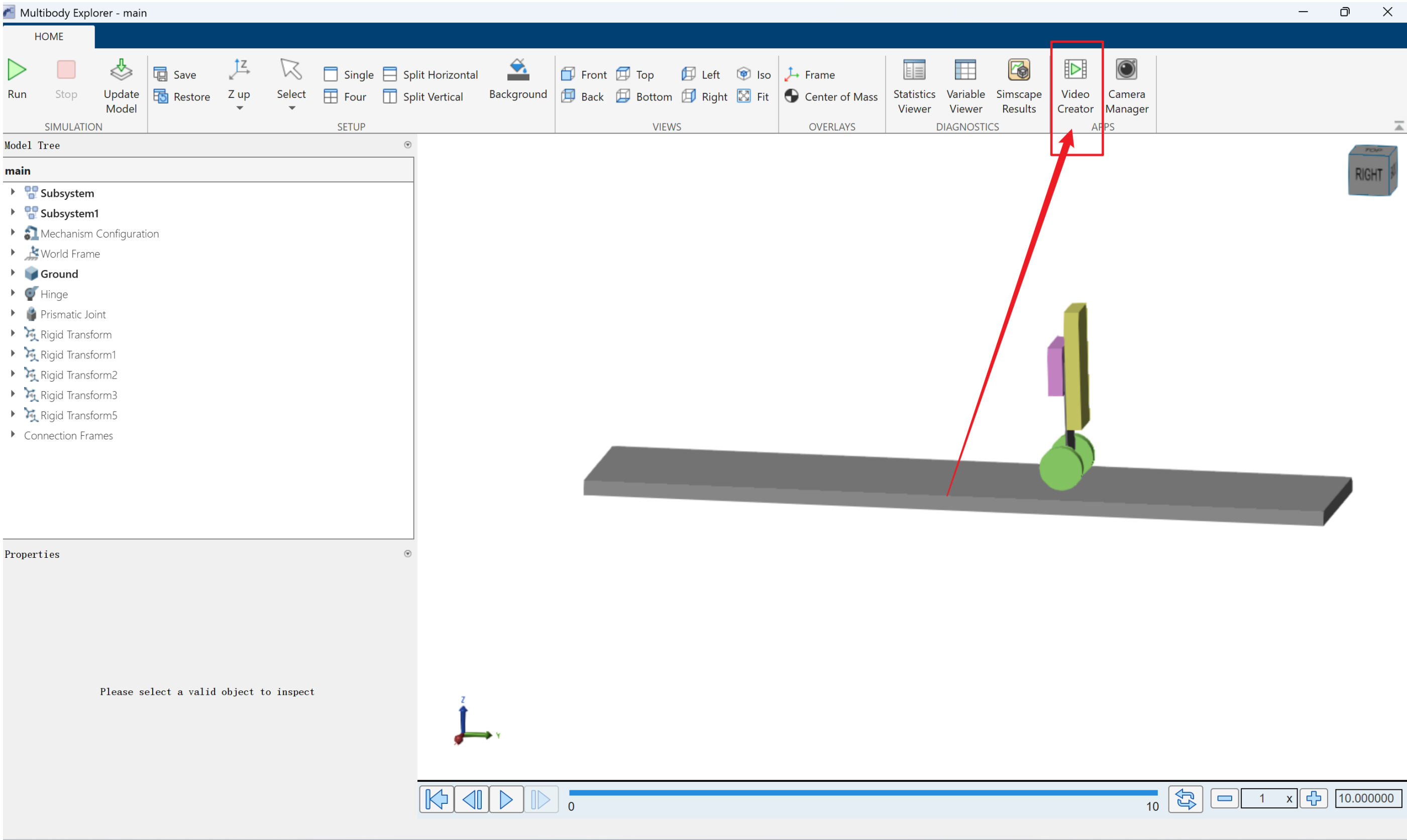
```
3 // Sampling frequency
4 #define SAMP_FREQ 2000.0f
5 // Motor voltage limit ( $V_{\text{motor}} = 3.3 * v_{\text{out}} * 4$ )
6 #define VLIMIT 0.1f
7 // Motor offset voltage
8 #define VOFFSET 0.0f
9 // Feedback gains (Arbitrary Unit)
10 #define KP 1.0f
11 #define KD 0.0f
12
```

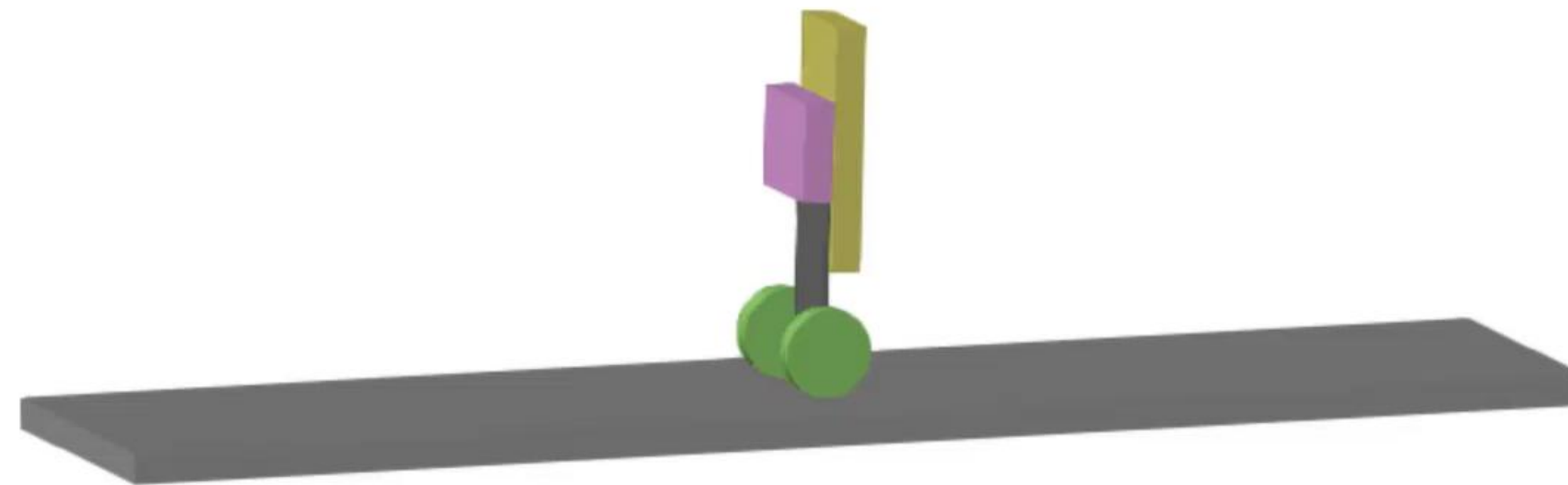


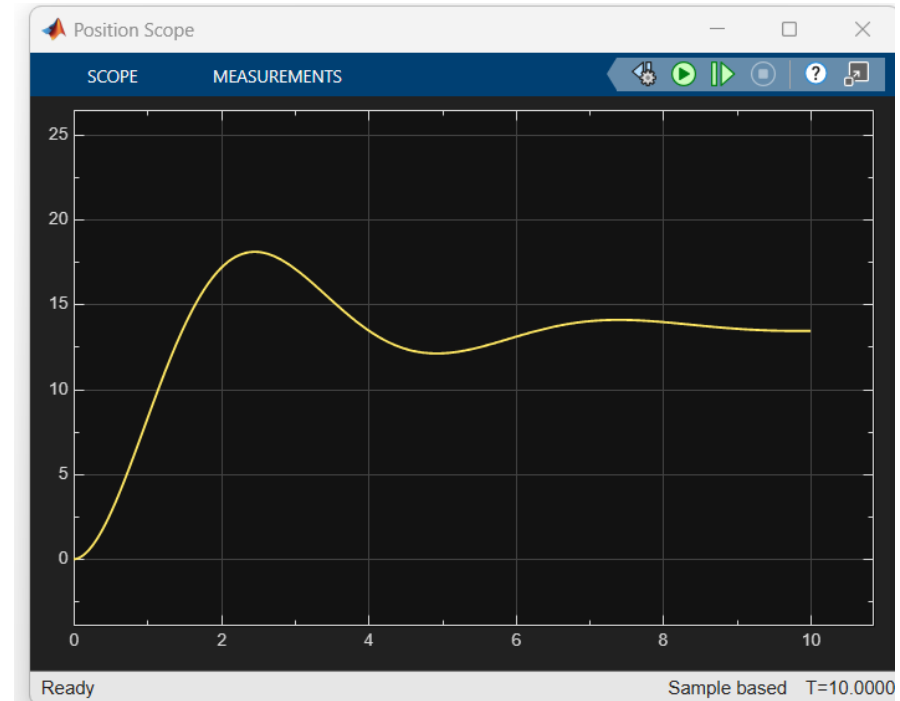
```
105 // Motor output
106 if (pd1v < 0.02f && pd2v < 0.02f) { // if not standing, stop the motor
107     motor_v = 0;
108     led_v_forward = 1;
109     led_v_backward = 1;
110 } else {
111     motor_v = v_out + VOFFSET;
112     led_v_forward = (v_dir > 0 ? v_out * 3.3f : 0.0f);
113     led_v_backward = (v_dir < 0 ? v_out * 3.3f : 0.0f);
114 }
115
```

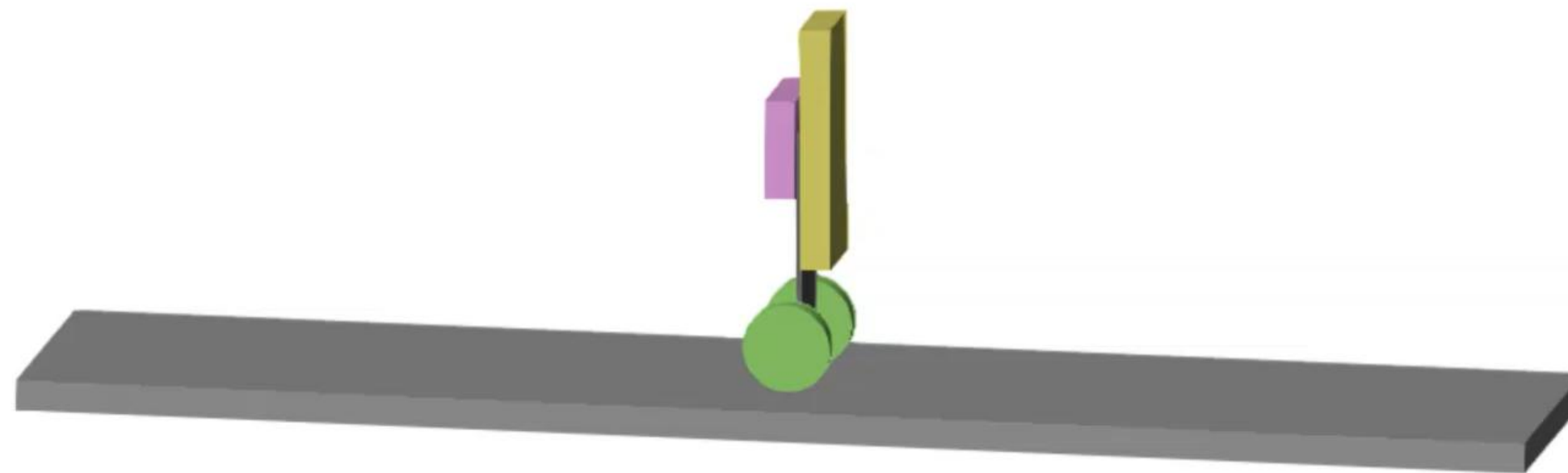
Use AI Tools to help you understanding the codes



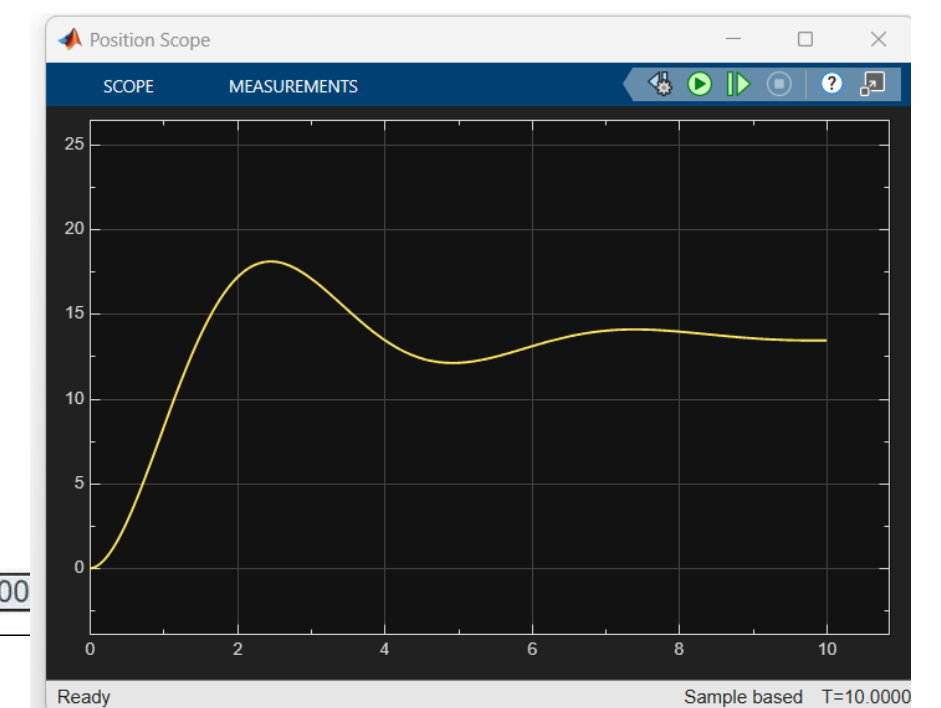


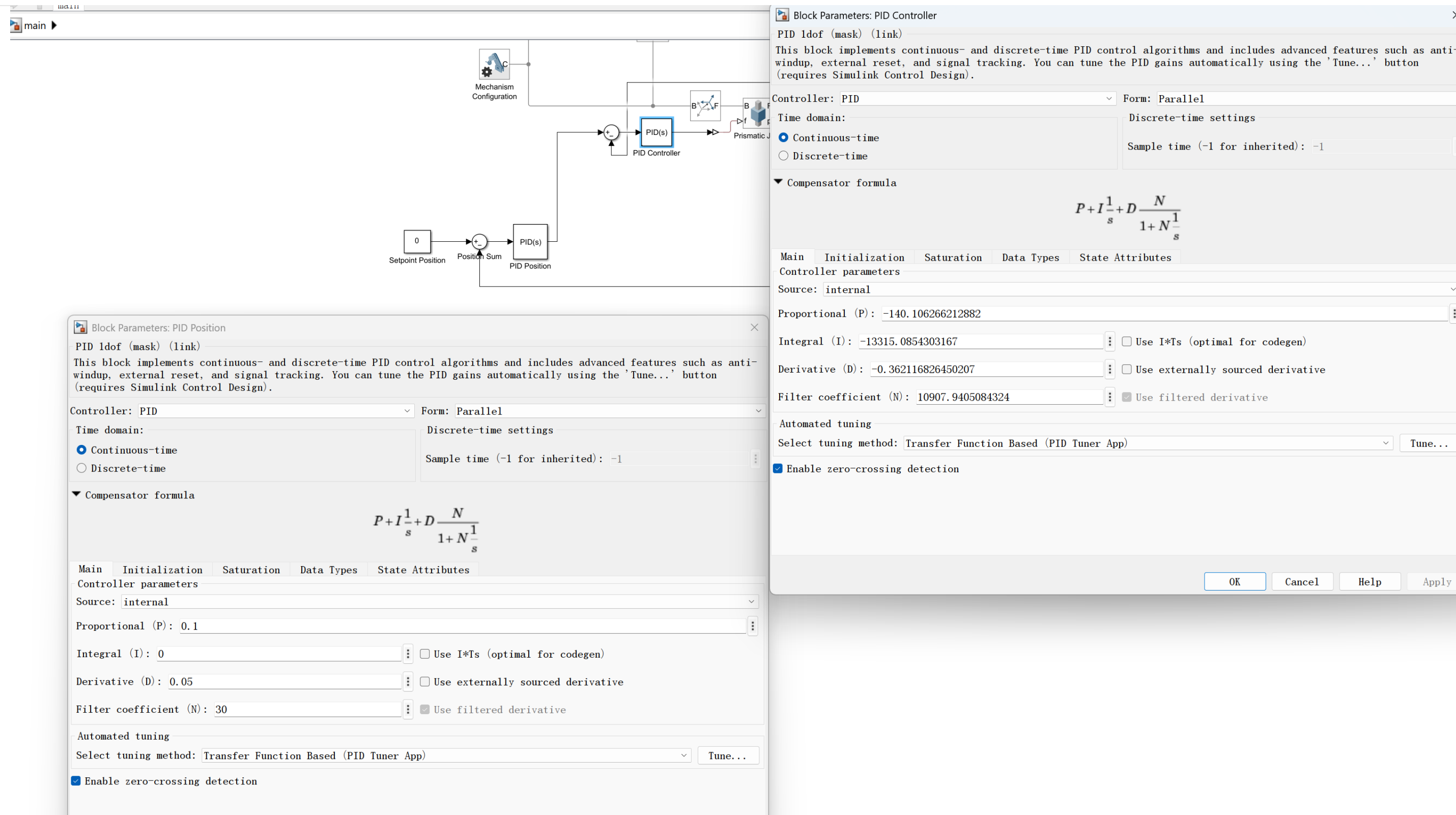






0.0000





Week 2 Simulink Simulation – Disturbance

13

Block Parameters: Hinge Disturbance Torque

Pulse Generator

Output pulses:

```
if (t >= PhaseDelay) && Pulse is on
    Y(t) = Amplitude
else
    Y(t) = 0
end
```

Pulse type determines the computational technique used.

Time-based is recommended for use with a variable step solver, while Sample-based is recommended for use with a fixed step solver or within a discrete portion of a model using a variable step solver.

Parameters

Pulse type: Time based

Time (t): Use simulation time

Amplitude: 0.05

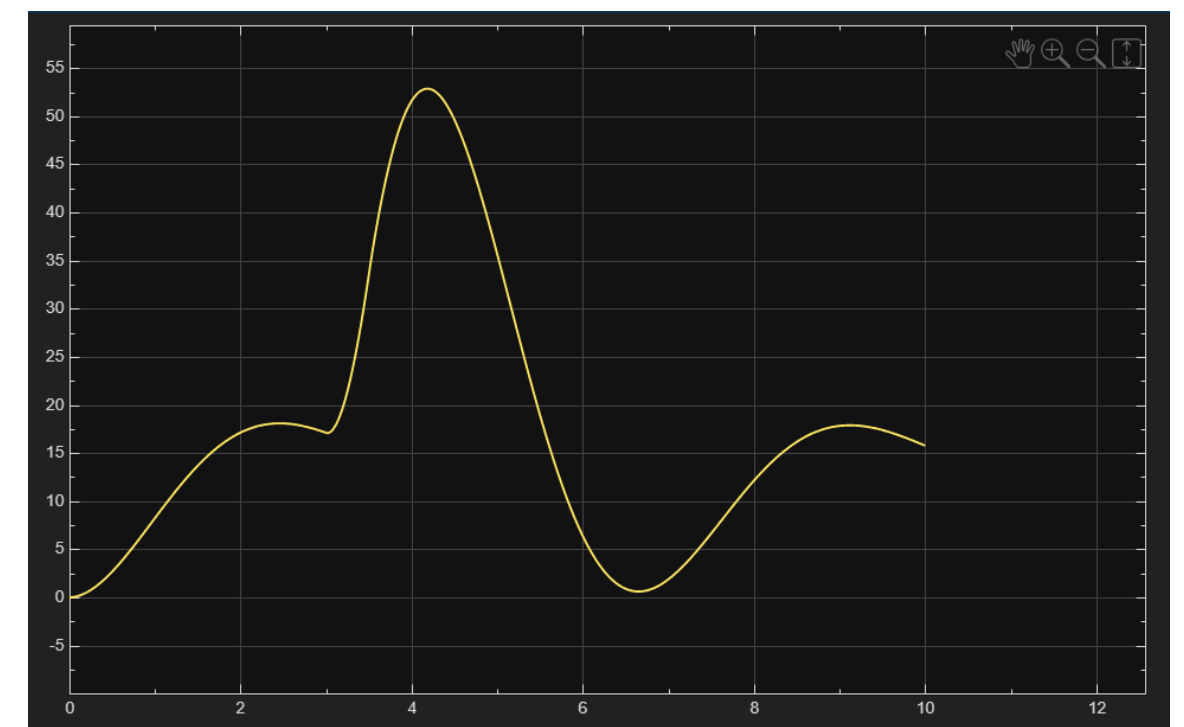
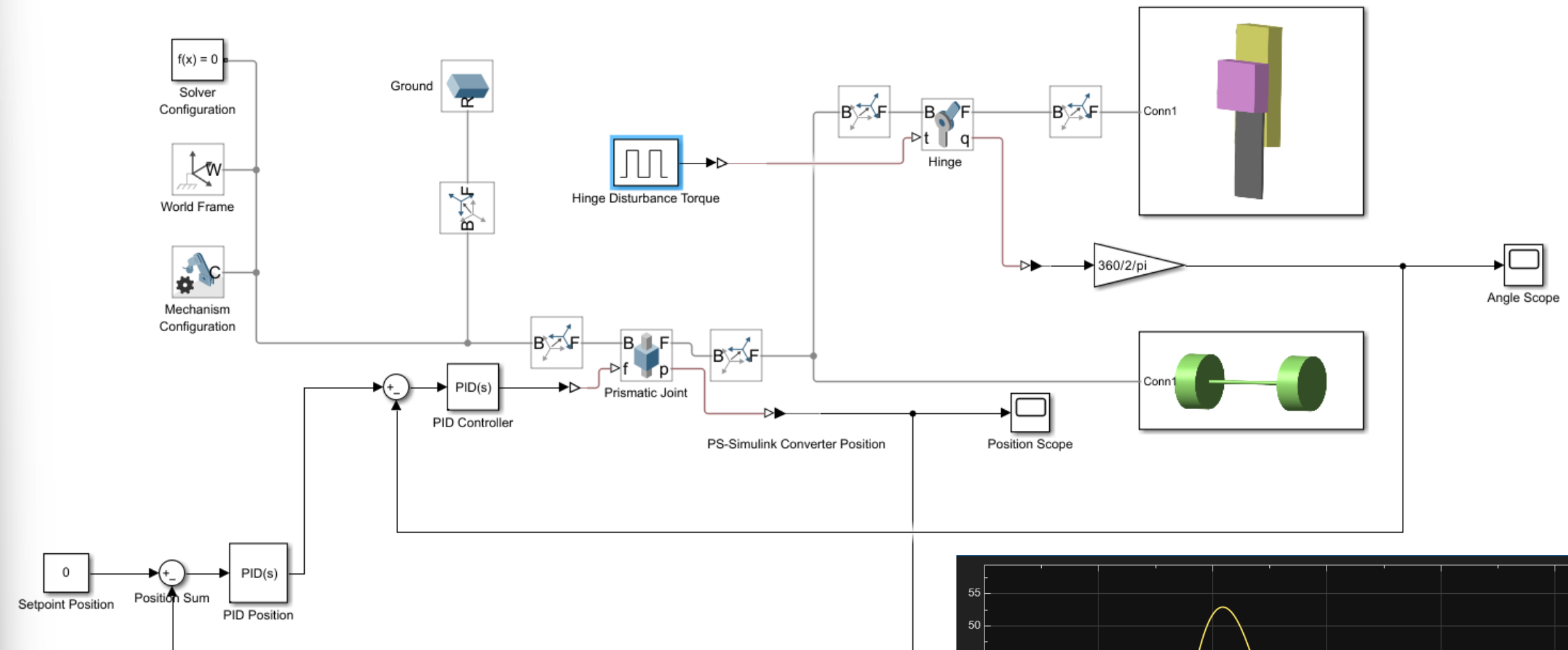
Period (secs): 10

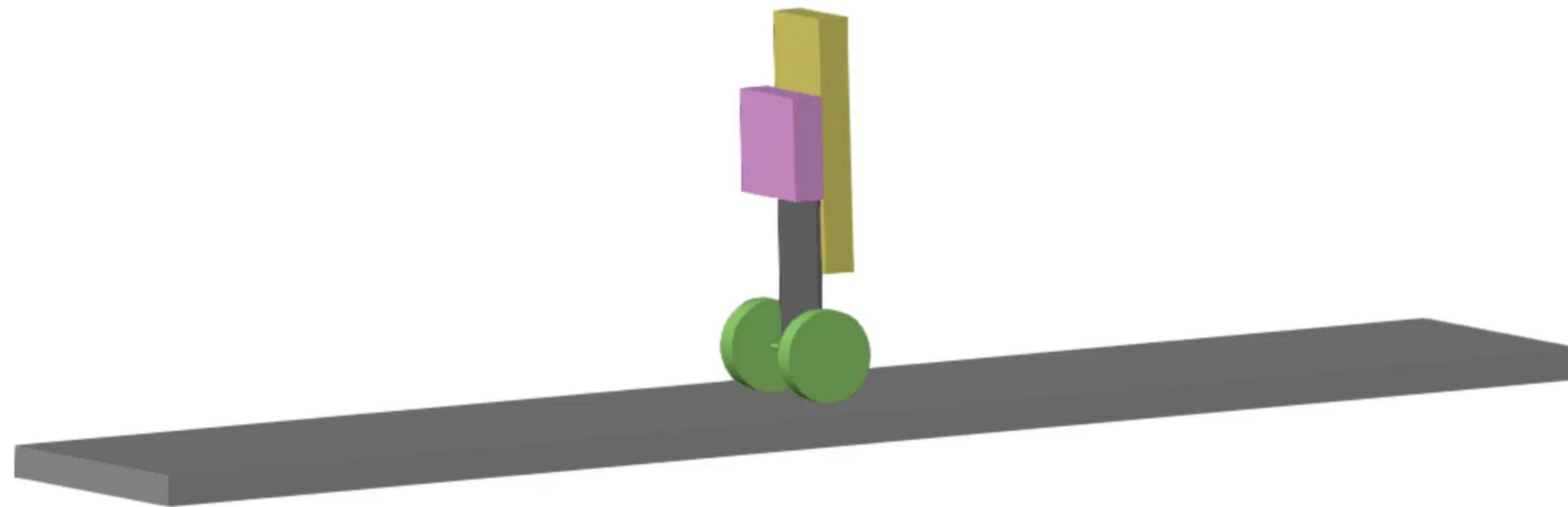
Pulse Width (% of period): 5

Phase delay (secs): 3

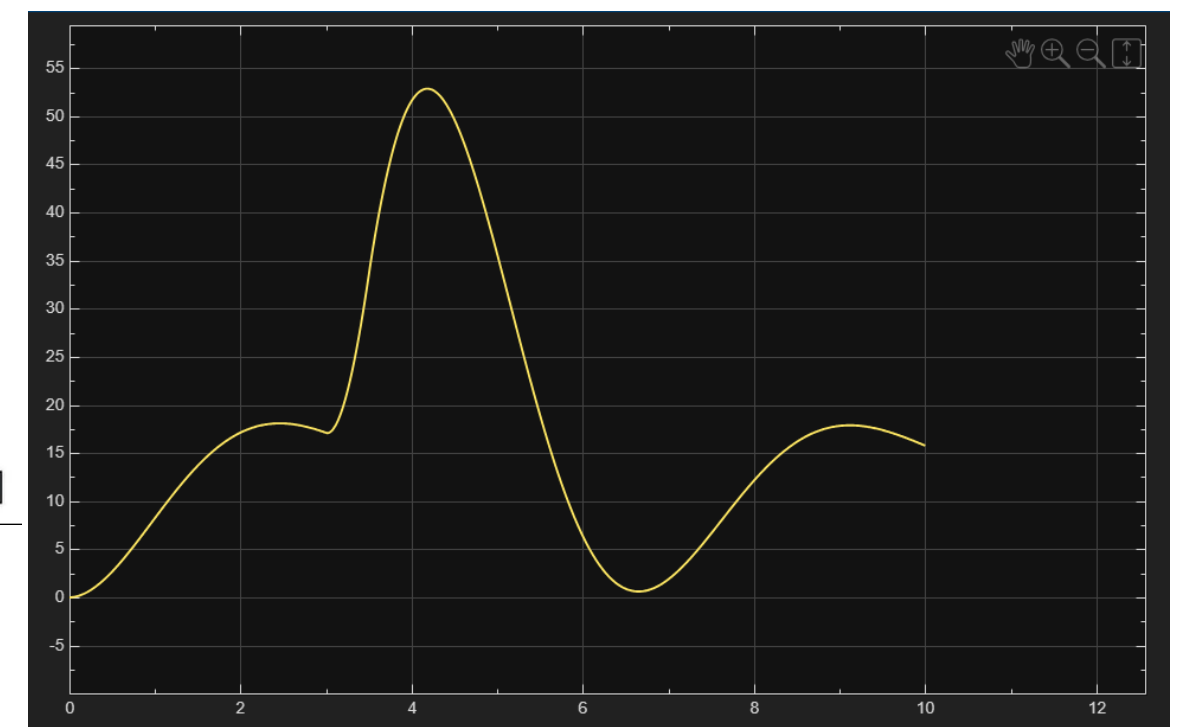
☒ Interpret vector parameters as 1-D

OK Cancel Help Apply





0.000000



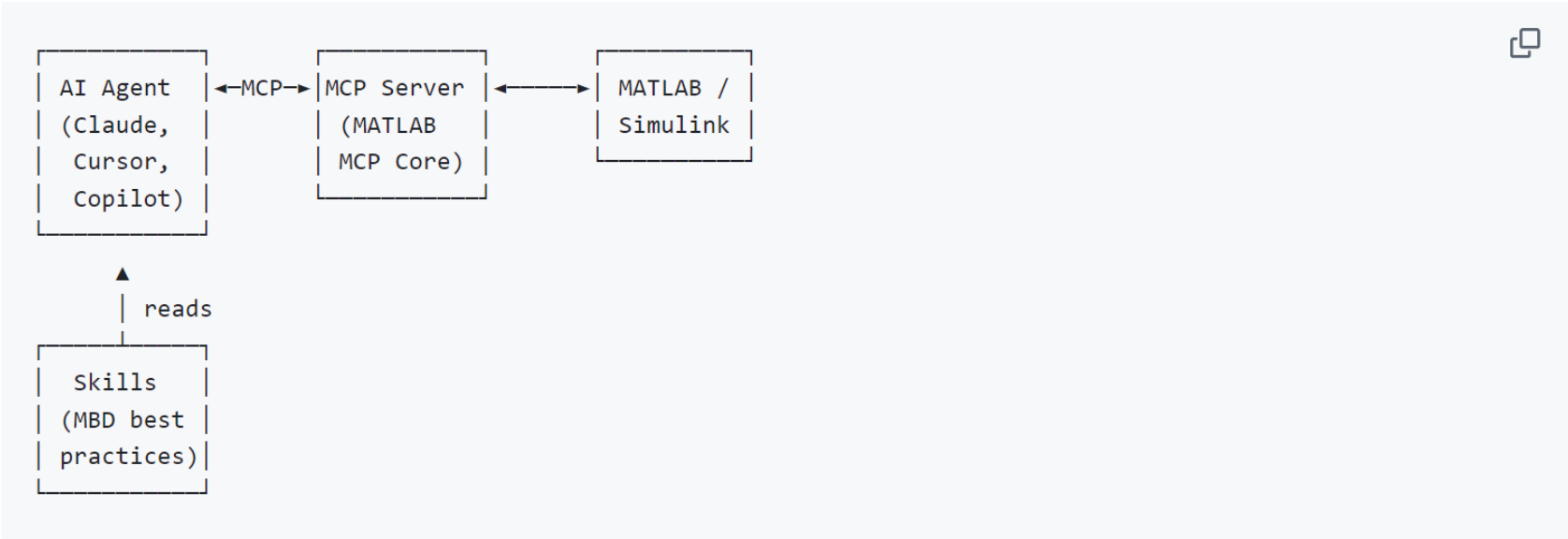
3.5.6 Running the Simulation

Locate the provided .SLX model file in the course material folder and open it in MATLAB. Before running, configure the simulation to match a realistic digital control loop. In **Simulation** → **Model Configuration Parameters** (Ctrl+E), use a **fixed-step** solver and choose a step size consistent with your intended control frequency (for example, 1 ms to match a 1 kHz loop, or another value if your real implementation uses a different rate). Matching the sample time is important because it affects stability margins and makes simulation results more predictive when you later transfer the controller to hardware.

To observe stabilization, set a small initial tilt angle in the Revolute Joint (for example, $\theta(0) = 0.1$ rad $\approx 5.7^\circ$) so the controller has a disturbance to reject. Then run the simulation and open the angle scope to check whether the angle converges back to the upright equilibrium. If the model includes a disturbance input, you can also inject a brief impulse or step disturbance and examine how quickly the controller recovers.

The fixed-step setting here may cause the simulation to fail. You can keep the adaptive step size instead.

How It Works



Your agent reads **skills** for domain knowledge, then calls **MCP tools** to interact with MATLAB and Simulink. The [MATLAB MCP Core Server](#) bridges the connection (downloaded during setup).

Supported Platforms

Platform	Setup	Notes
Claude Code	Automated	Also supports no-clone marketplace install (skills only)
GitHub Copilot	Automated	
OpenAI Codex	Automated	
Gemini CLI	Automated	
Sourcegraph Amp	Automated	
Cursor	Manual	Untested

<https://github.com/matlab/simulink-agentic-toolkit>

Command Window

```
matlab.addons.install('C:\Users\rocke\.local\share\MATLABMCPCoreServerToolkit.mltbx')
addpath('D:\Local\20251230_Inverted_Pendulum\Week2\simulink-agentic-toolkit')
satk_initialize

ans =

1x4 table

      Name      Version  Enabled  Identifier
      _____  _____  _____  _____
"MATLAB MCP Core Server Toolbox" "0.1.0"    true    "fce99a56-1b63-4d49-9d70-c32c5605e4eb"
```

Simulink Agentic Toolkit - Installation Check

=====

Prerequisites

- ✓ MATLAB 25.2.0.3055257 (R2025b) Update 2
- ✓ Simulink installed

MATLAB Environment

- ✓ All 6 tool entry points on path
- ✓ Package resolution correct (+sage, +mbd inside SATK)

MCP Server

- ✓ matlab-mcp-core-server.exe found: C:\Users\rocke\.local\bin\matlab-mcp-core-server.exe

MCP Connectivity

- ✓ shareMATLABSession available (MATLABMCPCoreServerToolkit installed)
- ✓ MATLAB connector running (port 31516)

=====

Result: PASS

Week 2 Submission

If the robot cannot work yet, record its current state and clearly show which parts are already working.

Please also attach a PDF explaining the current status and the planned solution.

Video 1:

show the real wheeled inverted pendulum cart under a disturbance.

Video 2:

show the simulation running under normal conditions without an external disturbance.

Video 3:

show the simulation response when a disturbance is applied.

DEADLINE

21:00

on the day of the class
Today — July 6

Late submissions accepted until **21:00 the following day** . Late score capped at 50%.

Each group only needs one submission.